

Two-Phase WESP Flush Chemistry Optimization

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CONTENTS



Title

Introduction

Internship Overview

Key Tasks

Project Impacts

Valuable Insights

Learning Experience

Acknowledgements

Conclusion

INTRODUCTION

About Me

- Honors College Senior
- University of South Carolina
- Chemical Engineering

About Enviva

- Prior exposure
- Biomass Energy
- Sustainability

About the Project

- Reduce WESP Buildup
- Financial Viability
- Conservation



INTERNSHIP OVERVIEW

Purpose

- Support engineering and operations at GRE
- Investigate opportunities to have the WESP:
 - Consume less caustic
 - Run more efficiently
 - Require less cleaning



Goals

- Analyze buildup causes
- Design solution
- Quantify monetary impact
- Identify operational benefits



Mission

“Displace fossil fuels, grow more trees, and fight climate change”



- Reduce maintenance
- Support consistent production
- Use less water/ energy



KEY TASKS – PROBLEM EVALUATION



WESP Fouling

- What causes the buildup?
- What is it made of?
- What does the buildup effect?

Site/Vendor Input

- Learning from [REDACTED]
- What [REDACTED] would change
- [REDACTED] advice

Flush Limitations

- Time-based intervals
- Operator error
- Blowdown issues

KEY TASKS – PROJECT DEVELOPMENT

Injection System

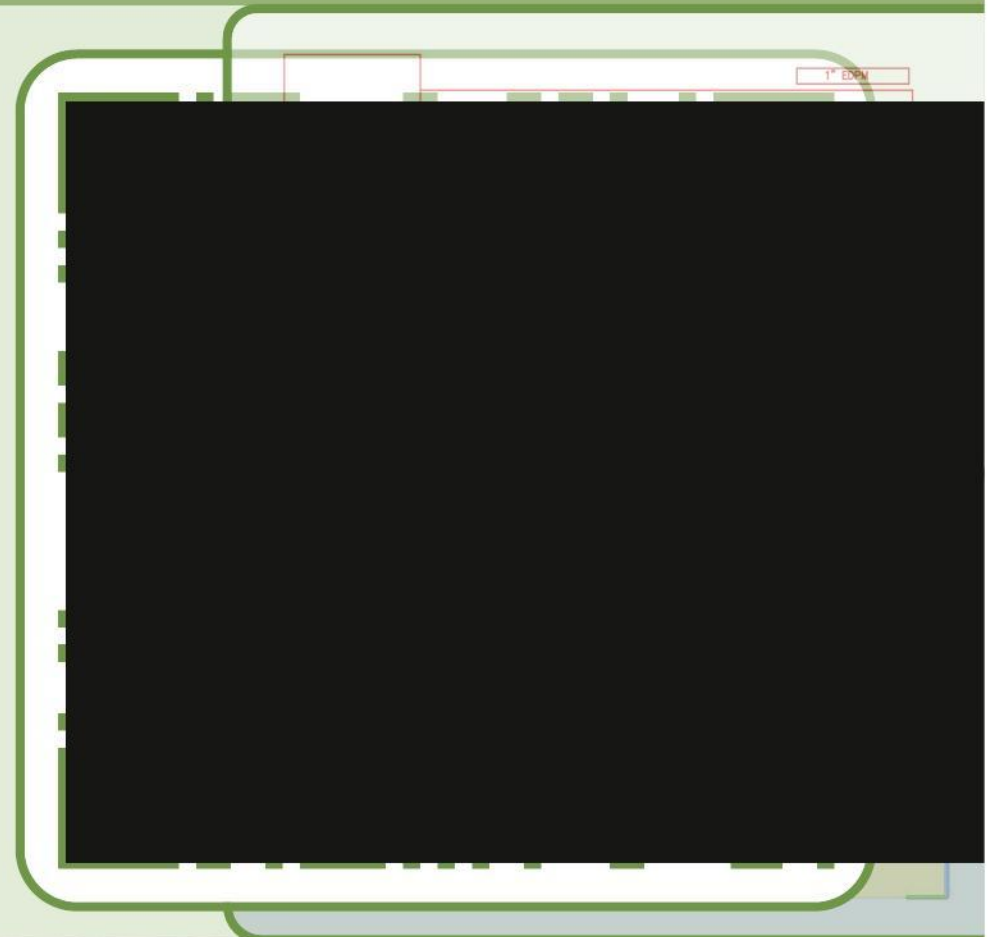
- Two phases:
 - Temporary
 - Permanent

Documentation

- Prediction Model
- PFDs/ P&IDs
- SOPs

Handoff

- Binder with:
 - Pipe cut lengths
 - Trial data collection
 - Phase 2 decision path



PROJECT IMPACTS

Before

- Buildup reduces throughput
- [REDACTED]/outage
- Longer outages

After

- Hands-off [REDACTED]
- Eliminate [REDACTED]
- Save valuable time

Economics

- NPV15 – [REDACTED]
- IRR – [REDACTED]
- Payback – [REDACTED]



VALUABLE INSIGHTS

Challenges

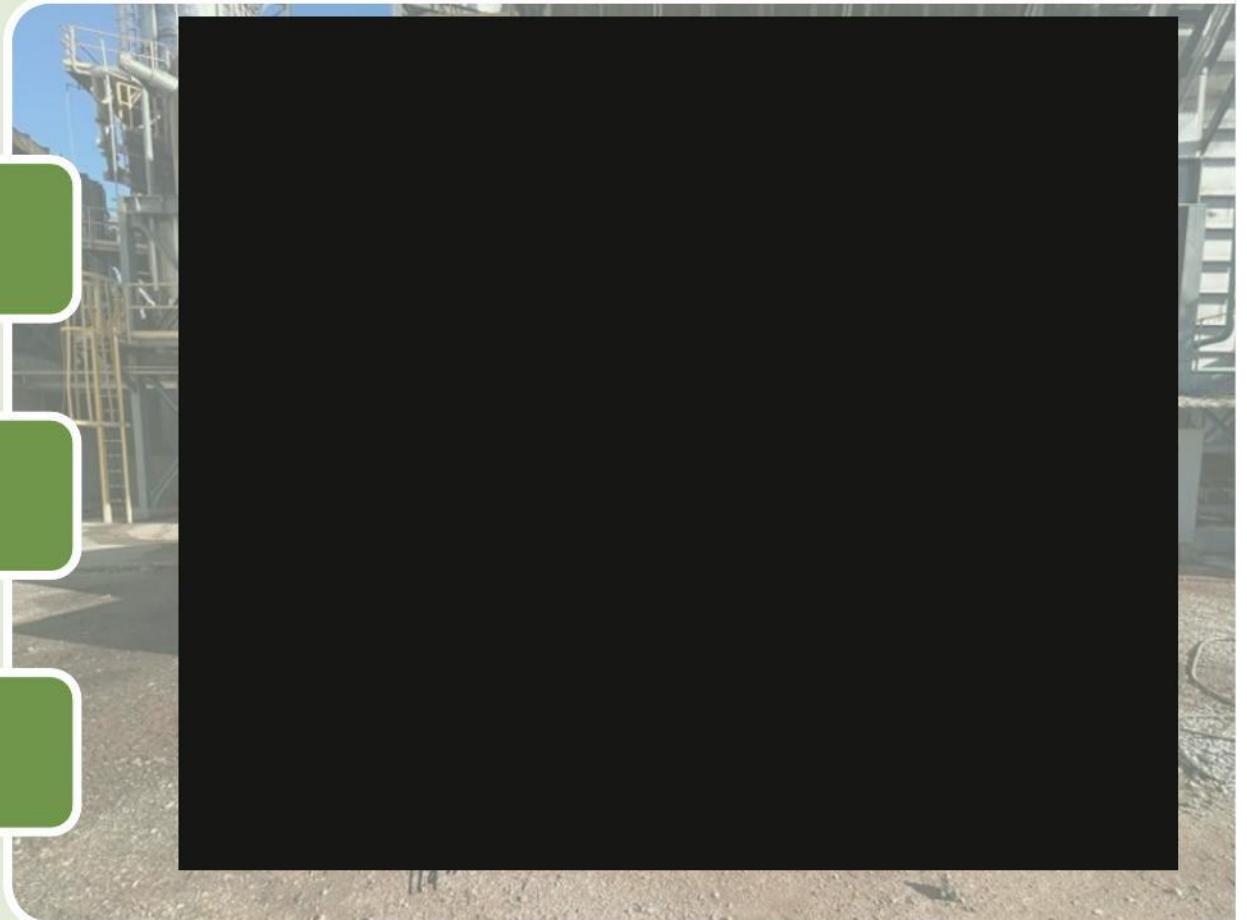
- Real plant constraints
 - Spatial
 - Mechanical

Lessons Learned

- Interdisciplinary feedback
- Diverse viewpoints
- Continually seek advice

Successes

- [REDACTED] connections
- Manager approval [REDACTED]
- Reached [REDACTED]

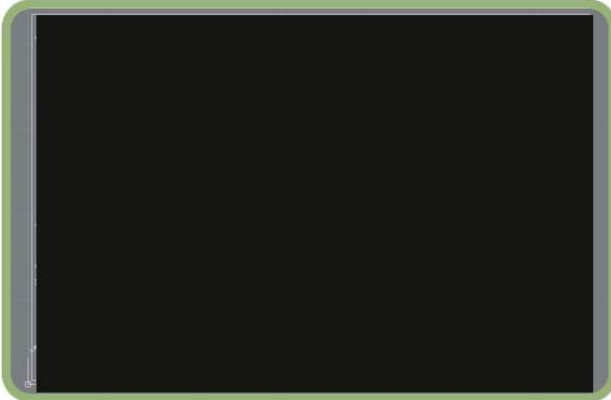


SKILLS DEVELOPED



Financial

- Excel cost breakdowns
- [REDACTED] file
- Quote comparisons and project estimating



Technical

- AutoCAD/ P&ID concepts
- Equipment layout understanding
- Piping and tie-in visualization



Professional

- Vendor communication
- Engineering feedback revisions
- Plant Safety and LOTO

LEARNING EXPERIENCE

Started with a question:

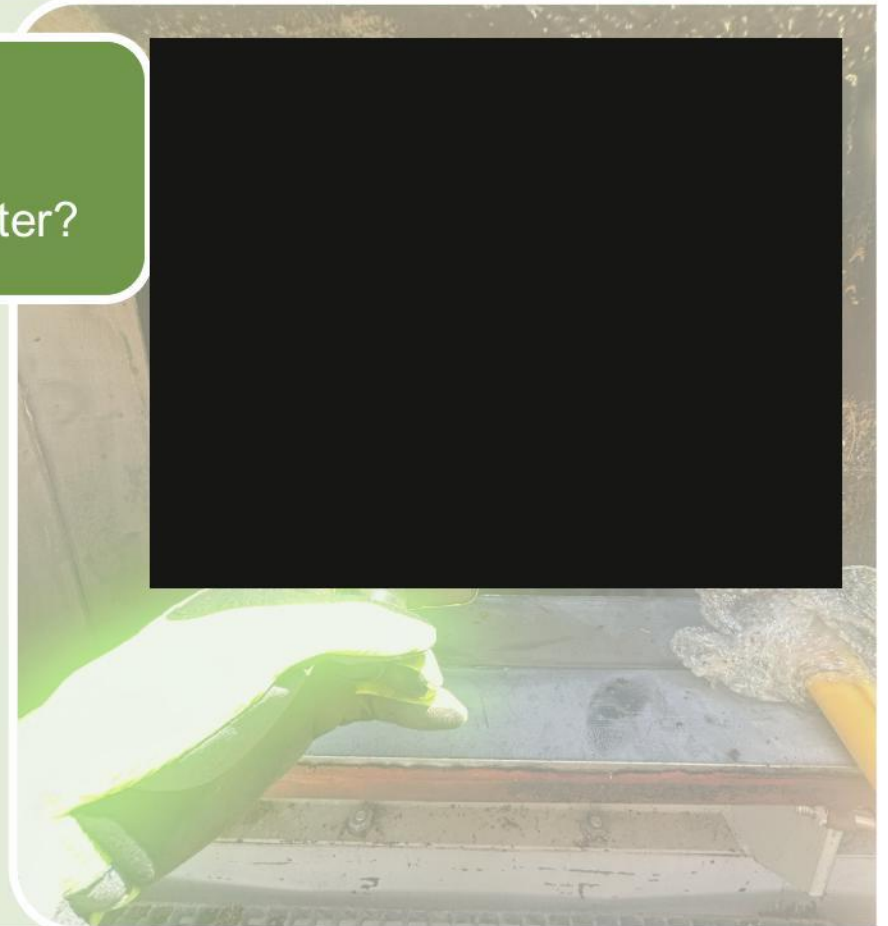
How can we make the WESP flush work better?

Pre-Internship

- Definitive problems and answers
- Limited financial exposure

Post-Internship

- Confidence in new tools
- Better understanding of engineering



SPECIAL THANKS

GRE

- Randy Moore
- Willie Middleton
- Jerry Padrta
- Luke Spach
- Andrea Bronson
- Matt Grimmatt



ETC

- Michael Hedgegaard (LUC)
- Bryan Barnett (COT)
- Stephen Ray (RelaDYNE)
- Intern Team
- Will Booth (RAL)



CONCLUSION

- Strengthened my understanding of real plant engineering
- Learned how to fully develop a design from start to finish
- Gained experience with AutoCAD, financial analysis, and plant safety
- Created something that will continue after my internship





Thank you!
Any Questions?

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